

Net Surfer

Eduard · Assignment 3 · a desktop web browser with its own HTML → DOM → CSS → layout → paint pipeline, in Python

ARCHITECTURE: HOW A URL BECOMES PIXELS



Fetching, parsing, styling and image decoding run on a worker thread; layout and painting run on the UI thread. **Libraries only for:** sockets/TLS/HTTP framing (`http.client`, `ssl`), window + text drawing (`tkinter`), raster image decoding (`Pillow`). Everything else is our code.

WHAT IT CAN DO

- HTTP/HTTPS, redirects, relative URLs, cookies, cache, gzip, charsets, GET/POST forms
- Forgiving HTML parser (implied tags, entities, bad nesting)
- CSS: external/inline sheets, `@import`, `@media`, cascade + specificity, `var()`, `calc()`, `::before`/`::after`
- Layout: box model, margin collapsing, inline text wrapping, inline-block, floats, lists, tables, flexbox, grid, absolute/relative positioning
- Paint: text, backgrounds, borders, PNG/JPEG/GIF images, SVG
- Tabs, back/forward, reload, scrolling, links, bookmarks, view-source, DOM/style inspector, network log

~7,000

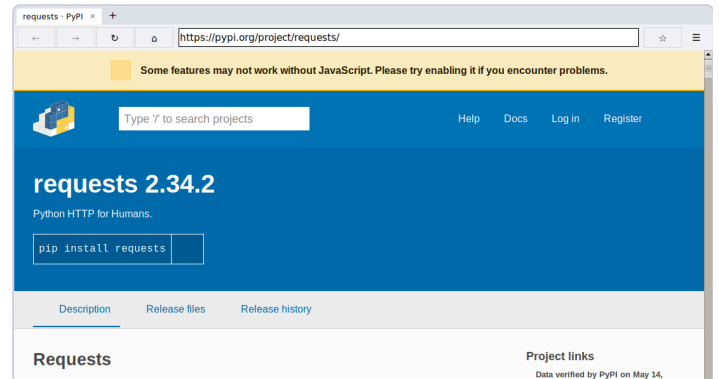
lines of our code

51

automated tests

0

crashes in fuzzing



pypi.org rendered by the browser (layout, flexbox, floats, clearfix, SVG logo)

MOST AMBITIOUS FEATURE

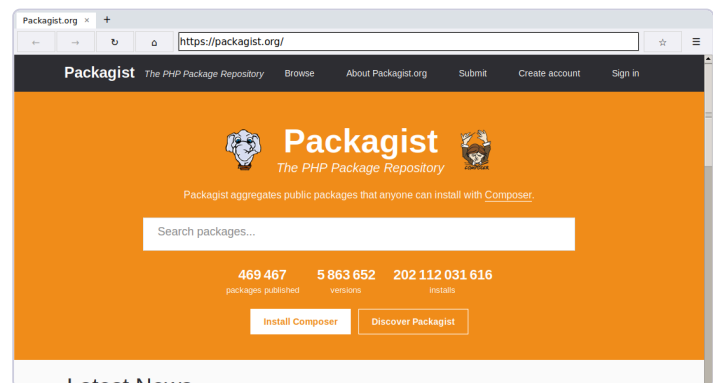
A real **layout engine** rather than a text dumper: one box tree handles block flow with margin collapsing, inline line boxes with baselines, floats that text wraps around, table column sizing from min/max content widths, flexbox grow/shrink, grid tracks and absolute positioning, all driven by computed styles. Plus our own **SVG rasterizer** (paths, arcs, transforms) so logos and icons show up.

MOST IMPORTANT LIMITATION

No JavaScript. Pages that build their content in JS (single-page apps) show only their static HTML. Also no web fonts (icons drawn with icon fonts are missing) and no z-index ordering. Documents nested thousands of levels deep show an error page because the layout is recursive.

HOW IT WAS TESTED

- Unit tests for the parser, selectors, cascade, URLs and layout geometry
- A local HTTP server for redirects, gzip, cookies, charsets, 404s, POST
- End-to-end tests that click links, go back/forward, type into and submit forms in the real window
- Random malformed HTML/CSS fuzzing through the whole pipeline
- Screenshots of real sites (PyPI, RubyGems, Packagist, pkg.go.dev, Node.js) compared with how they should look



packagist.org: flex header, centred hero, search form